UNIVERSITY OF ALBERTA

Engineering
Co-op Program

Mohamednur Zeinu

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Portfolio: https://mohamednurz.github.io/portfolio/

ACADEMIC & CO-OP STATUS

Electrical Engineering, BASc Co-op, University of Alberta	**Class of 2029**
Cumulative Grade Point Average	**3.2/4.0**
Completed Academic Terms	**4 of 8**
Completed Co-op Work Terms	**0 of 4**
Availability Starting January 2027	**8 months**

TECHNICAL SKILLS

EQUIPMENT	**DESIGN**	**PROGRAMMING**	**PRODUCTIVITY**
Oscilloscope	Altium Designer	Python	MS Office (Word, Excel & PowerPoint)
Soldering (SMT/THT)	KiCad	C / C++	Wave Form
Breadboarding	Vivado	Assembly (ARM)	STM32cubeMX
Hardware Debugging		System Verilog, VHDL	

PROJECT EXPERIENCE

16-Bit CPU Design — July 2026 – Aug. 2026
Personal Project

- Designed a custom 16-bit CPU instruction set architecture (ISA) spanning 16 instructions across 5 encoding formats, defining a complete register-based, load/store processor from scratch.
- Implemented 9 SystemVerilog hardware modules forming a full single-cycle data path, including a 4096-word instruction memory and 4096-word data memory with memory-mapped I/O.
- Programmed a two-pass Python assembler translating custom assembly into 16-bit machine code. Then verified full CPU functionality by executing a 12-instruction Fibonacci sequence program, confirming 8 computed values and final register states matched a hand-traced expected execution with zero errors.
- Collaborated with a project partner on hardware and board-level constraints, jointly reviewing and debugging synthesis, timing, and pin-mapping issues to prepare the design for physical FPGA deployment.
- Synthesized and deployed the complete processor onto a Xilinx Zynq-7000 FPGA, achieving ---- MHz clock frequency while utilizing --- % of available FPGA logic resources.

Flight Controller (Project Maverick) — May 2025 – July 2025
Personal Project

- Designed a custom drone flight-controller PCB from scratch in Altium Designer with schematic capture, component selection, and full PCB layout.
- Selected and placed 86 components across 47 unique BOM line items, creating 14 custom PCB footprints (5 built to IPC standard) for parts with no stock library match.
- Integrated a STM32F405RGT6 ARM Cortex-M4 MCU (168 MHz) with a BMI088 IMU, BMP388 barometer, magnetometer, EEPROM, NAND flash, USB, SBUS input, and PWM motor driver.
- Ran a Design Rule Check across 13 constraint categories on the completed 86-component layout, achieving 0 short-circuit violations and 100% net routing completion (0 unrouted nets).
- Built a real-time flight-stabilization control system in Python, fusing IMU accelerometer and gyroscope data with a complementary filter (98%/2% weighting) to estimate pitch/roll orientation at a 50 Hz loop rate.

ADDITIONAL WORK EXPERIENCE

Automotive Service Advisor — Sep 2025 – Present

Canadian Tire, Edmonton, AB

- Conduct structured vehicle diagnostic interviews and translate technical findings into clear service recommendations, coordinating repair orders between customers and licensed automotive technicians.
- Manag concurrent service desk operations including inbound calls, walk-in consultations, and multi-vehicle status tracking with more than 25 vehicles a day on average resulting in more than $300K+ average monthly revenue.
- Monitor repair timelines, proactively communicate vehicle status updates, and resolve service exceptions to maintain customer satisfaction and retention.

Senior Automotive Service Advisor Apr 2023 – Sep 2024
Canadian Tire, Fort McMurray, AB

- Facilitated 20+ daily customer service interactions by scheduling appointments, processing warranty claims worth $30K+ monthly, and coordinating with technicians to complete vehicle repairs on time.
- Produced detailed estimates for labor and parts, and coordinated with technicians to confirm diagnostics, repair progress, and completion timelines.
- Trained and mentored 3 new Service Advisors, providing onboarding support and technical guidance on shop management systems and customer handling procedures.

SCHOLARSHIPS & AWARDS

ATCO Centennial Award – ATCO 2025 – 2026
Jason Lang Scholarship – University of Alberta 2024 – 2025
Speaker of the Week 5x – UAPSSA 2024 – 2025

LEADERSHIP & COMMUNITY INVOLVEMENT

Member and Board (Nominee), CIIRSA – Fort McMurray 2023 – Present

ADDITIONAL INFORMATION

Willing to relocate
Interests – GYM Strength training, FPV drones, electronics design, Hip-Hop music, Playing Soccer, Cycling.